



BAULKHAM HILLS HIGH SCHOOL

2015

HSC Assessment Task 1

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 50 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work
- Attempt all questions.
Use the answer booklet provided.
- Answer each question on the appropriate page

Total marks –42

Exam consists of 4 pages.

This paper consists of TWO sections.

Section 1 – Page 2

Multiple Choice

Question 1-3 (3 marks)

Section 2 – Pages 3-5

Extended Response

Question 4-6 (39 marks)

Section I - 5 marks

Answer in the table provided in the answer booklet

1. What value of z satisfies $z^2 = 7 - 24i$?

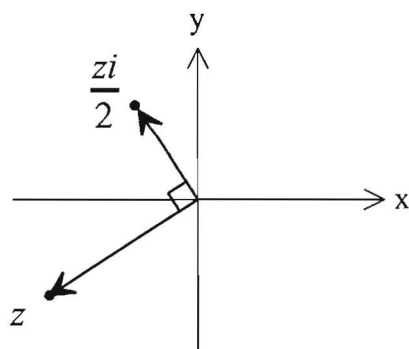
- (A) $3 - 4i$
- (B) $-3 - 4i$
- (C) $4 - 3i$
- (D) $-4 - 3i$

2. Which of the following is true for all complex numbers of z ?

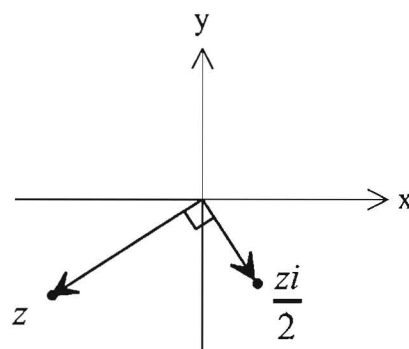
- (A) $\text{Im}(z) = \frac{z+\bar{z}}{2}$
- (B) $\text{Im}(z) = \frac{z-\bar{z}}{2}$
- (C) $\text{Im}(z) = \frac{z+\bar{z}}{2i}$
- (D) $\text{Im}(z) = \frac{z-\bar{z}}{2i}$

3. Which Argand diagram could show the complex numbers z and $\frac{zi}{2}$?

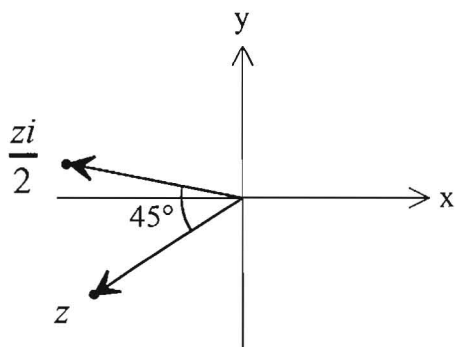
(A)



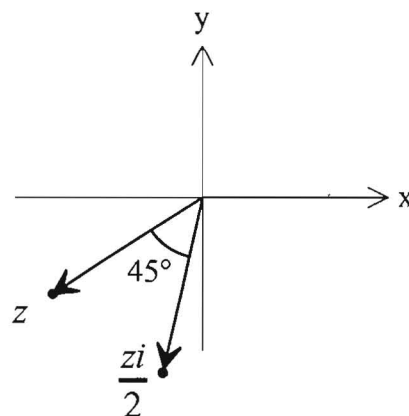
(B)



(C)



(D)



End of Section I

Section II – Extended Response

Attempt questions 6-10.

Answer each question on the appropriate page of the answer booklet, showing all necessary working.

Question 4 (12 marks)		Marks
(a) (i)	Express $1 + i$ in modulus argument form	2
(ii)	Hence evaluate $(1 + i)^{12}$	2
(b)	Let $z = 3 + i$ and $w = 1 - i$. Find, in the form $x + iy$,	
(i)	$2z + iw$	2
(ii)	$\overline{zw}$	2
(iii)	$\frac{6}{w}$	2
(c)	Find real numbers x and y such that $(1 + i)x + (2 - 3i)y = 10$	2

End of Question 4

Question 5 (13 marks)

a)	On an Argand diagram, the points P, Q, R represent the complex roots $1, w$ and w^2 . If $1, w$ and w^2 are the 3 roots of $z^3 - 1 = 0$. Find the area of the triangle ΔPQR .	3
b)	Let ω be one of the non-real roots of the equation $z^3 + 27 = 0$	
(i)	Factorise the cubic polynomial $z^3 + 27$ over the field of real numbers.	1
(ii)	Show that $\omega^2 - 3\omega + 9 = 0$	1
(iii)	Hence find the value of $\left(\frac{\omega^2}{3} + 3\right)^6$	3
c)	Sketch the following loci on separate argand diagrams	
(i)	$\operatorname{Re}\left(\frac{z-4}{z}\right) = 0$	3
(ii)	$\frac{3\pi}{4} \leq \arg(z - 1) \leq \pi$ and $ z - 1 - i \leq 1$	3

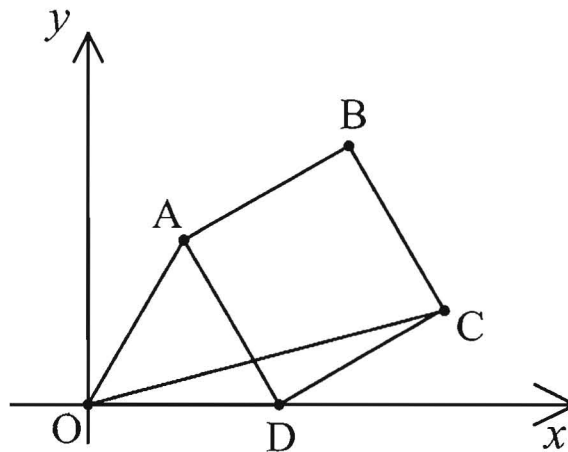
End of Question 5

Question 6 (14 marks)**Marks**

- a) It is given that $ABCD$ is a square and ABO is an equilateral triangle.

3

If the vector $\overrightarrow{CB} = -1 + i\sqrt{3}$, find the complex number represented by the point C .



- b) (i) Solve for $z^4 = 1$ for all z

1

- (ii) Hence or otherwise, solve $z^4 = (z - 1)^4$

3

- c) (i) Show that the roots $z^5 + 1 = 0$ on a unit circle in an Argand diagram.

2

- (ii) Factor $z^5 + 1$ into irreducible quadratic and linear factors with real coefficients

2

- (iii) Deduce that $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$ and $\cos \frac{\pi}{5} \cos \frac{3\pi}{5} = -\frac{1}{4}$

2

- (iv) Write a quadratic equations with integer coefficients which has roots $\cos \frac{\pi}{5}$ and $\cos \frac{3\pi}{5}$.

1**End of Exam**

BAULKHAM HILLS HIGH SCHOOL
EXTENSION 1 TRIAL HSC 2017 SOLUTIONS

Solution	Marks	Comments
<i>Section 1</i>		
1 C - By expanding looking $2abi$ term and then $a^2 - b^2$	1	
2 D - $\frac{x+iy-(x-iy)}{2i}$	1	
3 B - $z \times \frac{i}{2} = z \times i \times \frac{1}{2}$, Multiplying by i rotates counter clockwise 90° Multiplying by $\frac{1}{2}$, halves the modulus	1	

Solution	Marks	Comments
QUESTION 4		
8a (i) $ z = \sqrt{1^2 + 1^2} = \sqrt{2}$ $\arg(z) = \tan^{-1}\left(\frac{1}{1}\right) = 45^\circ \text{ or } \frac{\pi}{4}$ $\therefore 1 + i = \sqrt{2} \text{ cis } 45^\circ$	2	2 marks • equivalent correct expressions 1 mark • correct modulus • correct argument
8a (ii) $(1 + i)^{12} = (\sqrt{2} \text{ cis } 45^\circ)^{12}$ $= 64 \text{ cis } 180^\circ$ $= -64$	2	2 marks • correct value 1 mark • correct use of DeMoirves Theorem • simplified to a correct principle argument
8b (i) $2z + iw = 2(3 + i) + i(1 - i)$ $= 6 + 2i + i - i^2$ $= 6 + 2i + i - i^2$ $= 7 + 3i$	2	2 marks • correct answer 1 mark • correctly simplifies $i^2 = -1$ • correct expansion and simplifies
8b (ii) $(3 - i)(1 - i) = 3 - 3i - i + i^2$ $= 2 - 4i$	1	2 marks • correct answer 1 mark • correctly simplifies $i^2 = -1$ • correct expansion and simplifies
8c $(1 + i)x + (2 - 3i)y = 10$ $x + xi + 2y - 3i = 10$ Equating Real: $x + 2y = 10 \dots\dots(1)$ Equating Imaginary: $x - 3y = 0$ $x = 3y \dots\dots(2)$ Sub (2) into (1): $3y + 2y = 10$ $y = 2$ Sub $y \rightarrow (2)$: $x = 6$ $\therefore x = 6, y = 2$	2	2 marks • correct answer 1 mark • Equates Real • Equates imaginary

Solution		Marks	Comments
QUESTION # 5			
7a	Method 1: Since the 3 roots are equally spaced then the angle between each root is 120°. $\text{Area} = 3 \left(\frac{1}{2} \times 1 \times 1 \times \sin 120^\circ \right)$ $\text{Area} = \frac{3\sqrt{3}}{4}$	2	3 marks • Correct solution 2 mark • Significant progress towards solution with good reasoning 1 mark • Progress towards solution with good reasoning
7b (i)	$z^3 + 27 = 0$ $(z + 3)(z^2 - 3z + 9) = 0$	1	1 mark • Correct solution
7b (ii)	since ω is a root of $z^3 + 27 = 0$ then $\omega^3 + 27 = 0$ $(\omega + 3)(\omega^2 - 3\omega + 9) = 0$ Since ω is non-real, then $\omega \neq -3$ $\omega^2 - 3\omega + 9 = 0$ $\omega^2 + 9 = 3\omega$	1	1 mark • Correct solution
7b (iii)	$\omega^2 + 9 = 3\omega$ $\frac{\omega^2}{3} + 3 = \omega \quad (\text{divided both sides by 3})$ $\left(\frac{\omega^2}{3} + 3 \right)^6 = \omega^6 \quad (\text{power of 6 on both sides})$ $\left(\frac{\omega^2}{3} + 3 \right)^6 = (\omega^3)^2$ $= (-27)^2 \quad (\text{since } \omega^3 = -27)$ $= 729$	3	3 marks • Correct solution 2 mark • Manipulates • Equates imaginary 1 mark • Correctly simplifies $\omega^3 = -27$ • Attempts to manipulate $\omega^2 - 3\omega + 9 = 0$
7c	$ CB = \sqrt{1^2 + \sqrt{3}^2} = 2$ $AB = BC = CD = DA$ (sides of a square are equal) $AD = DO = OA$ (sides of an equilateral triangle are equal) $\therefore OD = 2$ $\overrightarrow{DC} = \overrightarrow{CB} \times -i$ $\overrightarrow{DC} = -i(-1 + i\sqrt{3})$ $\overrightarrow{DC} = \sqrt{3} + i$ $\overrightarrow{OC} = \overrightarrow{OD} + \overrightarrow{DC}$ $\overrightarrow{OC} = 2 + \sqrt{3} + i$	3	3 marks • Correct solution 2 mark • Significant progress towards solution with good reasoning 1 mark • Finds modulus of CB • Uses i to rotate to the correct direction

QUESTION # 6			
8a (i)	Let $z = x + iy$ $\text{Re} \left(\frac{x + iy - 4}{x + iy} \right) = 0$ $\text{Re} \left(\frac{(x - 4 + iy)(x - iy)}{(x + iy)(x - iy)} \right) = 0$ $\text{Re} \left(\frac{x(x - 4) + y^2 + i(yx + y(x - 4))}{x^2 + y^2} \right) = 0$ $\frac{x^2 - 4x + y^2}{x^2 + y^2} = 0$ $(x - 2)^2 + y^2 = 4$ $\therefore$ it is a circle with centre (0,2) and radius 2	2	2 mark • Correct solution and diagram 1 mark • Progress towards solution with good reasoning • Uses $z = x + iy$ correctly • Extracts the real part of an expression

8a (ii)		3	3 mark - Correct solution showing open circle and showing intersection at (1,0) and (0,1) 2 mark - Understands both loci 1 mark - Draws of the loci correctly.
8b (i)	$z = \pm 1, \pm i$	1	1 mark Correct solution
8b (ii)	$z^4 = (z-1)^4$ $\frac{z^4}{(z-1)^4} = 1$ $\left(\frac{z}{z-1}\right)^4 = 1$ $\frac{z}{z-1} = \pm 1$ $z = \pm(z-1)$ $z = +(z-1) \quad \left \quad z = -(z-1) \right.$ $2z = +1 \quad \left \quad 2z = +1 \right.$ $z = \frac{1}{2} \quad \left \quad z = \frac{1}{2} \right.$ $\frac{z}{z-1} = \pm i$ $z = +iz - i \quad \left \quad z = -iz + i \right.$ $z(1-i) = -i \quad \left \quad z(1+i) = i \right.$ $z = \frac{-i}{1-i} \quad \left \quad z = \frac{i}{1+i} \right.$ $z = \frac{1-i}{2} \quad \left \quad z = \frac{1+i}{2} \right.$ (or just realising that real coefficients will give roots in conjugate pairs) $\therefore z = \frac{1}{2}, \frac{-1 \pm i}{2}$ (Note: there are only 3 roots as when you expand z^4 will cancel out)	3	3 mark - Correct solution 2 mark - Has 3 correct equations to solve for z. 1 mark - Manipulates to an appropriate equation so solve. - Solves for $\frac{z}{z-1}$ or $\frac{z-1}{z}$
8c (i)	$z^5 = -1, z^5 = \text{cis}(\pi + 2n\pi), \text{ where } n = 0, 1, 2, 3, 4$ $z = \text{cis}\left(\frac{\pi}{5} + \frac{2n\pi}{5}\right), z_0 = \text{cis}\frac{\pi}{5}$ $z_1 = \text{cis}\frac{3\pi}{5}, z_2 = \text{cis}\pi = -1, z_3 = \text{cis}\frac{7\pi}{5} = \text{cis}\left(\frac{-3\pi}{5}\right),$ $z_4 = \text{cis}\frac{9\pi}{5} = \text{cis}\left(-\frac{\pi}{5}\right)$	2	2 marks - correct solution 1 mark - Finds roots only. - Draws diagram only

8c (ii)	$z^5 + 1 = (z - z_0)(z - z_1)(z - z_2)(z - z_3)(z - z_4)$ $= [(z - z_0)(z - z_4)][(z - z_1)(z - z_3)](z + 1)$ $z_0 + z_4 = \operatorname{cis} \frac{\pi}{5} + \operatorname{cis} \frac{-\pi}{5} = 2 \cos \frac{\pi}{5}$ $z_1 z_4 = 1,$ $z_1 + z_3 = \operatorname{cis} \frac{3\pi}{5} + \operatorname{cis} \left(\frac{-3\pi}{5} \right) = 2 \cos \frac{3\pi}{5},$ $z_1 z_3 = 1$ $z^5 + 1 = (z + 1) \left(z^2 - 2z \cos \frac{\pi}{5} + 1 \right) \left(z^2 - 2z \cos \frac{3\pi}{5} + 1 \right)$	2	2 marks • correct solution 1 mark • Correctly simplifies $z_1 + z_2$. Correctly simplifies $z_1 z_2$.
8c (iii)	Substitute $z = i$: $(i + 1) = (i + 1) \left(-2i \cos \frac{\pi}{5} \right) \left(-2i \cos \frac{3\pi}{5} \right)$ $1 = -4 \cos \frac{\pi}{5} \cos \frac{3\pi}{5}$ $\cos \frac{\pi}{5} \cos \frac{3\pi}{5} = -\frac{1}{4}$ Substitute $z = 1$: $2 = 2(2 - 2 \cos \frac{\pi}{5})(2 - 2 \cos \frac{3\pi}{5})$ $\frac{1}{4} = (1 - \cos \frac{\pi}{5})(1 - \cos \frac{3\pi}{5})$ $\frac{1}{4} = 1 - (\cos \frac{\pi}{5} + \cos \frac{3\pi}{5}) + \cos \frac{\pi}{5} \cos \frac{3\pi}{5}$ $\frac{1}{4} = 1 - (\cos \frac{\pi}{5} + \cos \frac{3\pi}{5}) - \frac{1}{4}$ $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$	2	2 marks • correct solution 1 mark • Proves one of the identities.
8c (iv)	$\text{sum of roots} = \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$ $\text{product of roots} = \cos \frac{\pi}{5} \cos \frac{3\pi}{5} = -\frac{1}{4}$ $x^2 - \frac{b}{a}x + \frac{c}{a} = 0$ $x^2 - \frac{1}{2}x - \frac{1}{4} = 0,$ $4x^2 - 2x - 1 = 0$	1	1 marks • correct solution